

$P = 0.219$ ). MVA found shorter distance to fovea ( $OR = 0.95$ ,  $P = 0.004$ ), post-operative subretinal fluid/exudative RD ( $OR = 3.14$ ,  $P = 0.006$ ) and vitreous hemorrhage ( $OR = 4.50$ ,  $P = 0.003$ ) are associated with increased RTT.

**Conclusion:** These data suggest that shorter distance to fovea and post-operative complications such as subretinal fluid/exudative RD and vitreous hemorrhage are associated with increased risk of RTT. This study can serve as prognostic tools to better counsel patients on the risks and benefits of brachytherapy and allow for early intervention in the management of RTT. These data need to be validated in a prospective trial investigating post-operative complications and subsequent RTT after the treatment of posterior uveal melanomas.

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### Choroidal Melanoma With Ultrasound-Guided Episcleral Brachytherapy: Long-Term Results and Risk of Metastasis

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**Purpose/Objective(s):** To elucidate the long-term outcomes in patients with choroidal melanoma who received episcleral brachytherapy with 125-I seeds; analyze cause-specific survival, metastasis-free survival and local control and establish the relationship between tumor size and metastases.

**Materials/Methods:** From May 2007 to February 2013, 88 patients classified according to American Joint Committee on Cancer guidelines underwent ultrasound-guided episcleral brachytherapy with a total prescribed dose of 72.40 Gy to the apex.

**Results:** Among the included cases, 47.7% and 44.3% had a clinical tumor stage of T2 and T3, respectively. With a median follow-up of 84 (range 7–153) months, local control at 5 and 10 years was 100% and 95%, respectively. Among the 88 patients, 9 (10.2%) were enucleated after brachytherapy. Those with T1–T2 and T3–T4 disease had a 10-year cause-specific survival of 100% and 87.3%, respectively ( $P = 0.017$ ). Metastasis-free survival at 5 and 10 years was 100% in those with T1–T2 disease and 92.1% and 83.1% in those with T3–T4, respectively ( $P = 0.016$ ). Five patients had liver metastases, all of whom had T3–T4 disease.

**Conclusion:** Ultrasound-guided episcleral brachytherapy with 125-I seeds yielded excellent local control for choroidal melanoma, with low complication rates and 90% eye preservation. Given the association between tumor stage and liver metastases, which remain the main cause of death, stricter control should be employed for T3–T4 tumors for the early detection and treatment of relapses.

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### Circumferential Stereotactic Body Radiotherapy for Spine Metastasis: The Feasibility, Toxicity and Local Control

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**Purpose/Objective(s):** Circumferential spine stereotactic body radiotherapy (CsSBRT) is indicated in selected patients with circumferential vertebral bone/epidural spine metastatic disease. There is a paucity of data on the feasibility, toxicity and outcomes of CsSBRT. This study aimed to determine the feasibility, toxicities and outcomes of CsSBRT treatments.

**Materials/Methods:** We analyzed sixty-four circumferential treatments from 2013-2019 treated with SBRT in the post-operative and intact spine metastatic setting. Radiotherapy was delivered with 1-5 fractions. We analyzed the local control, side effects and the dose-volume histogram for each treatment.

**Results:** In the sixty-four plans, the median follow-up was 15 months (1-63 months). All images and treatment plans were reviewed by the CNS radiation oncology and neuro-radiology. The most common regimen used was 27 Gy in 3 fractions. The 12-month cumulative incidence of local failure was 15%. In this cohort, there was eight local failures; 5 of them had recurrence in the epidural space, two had disease progression in the vertebral body and one had disease progression in the vertebral body with extension to the right lateral compartment. The planning target volume (PTV) coverage was  $\geq 95\%$  in 60 plans (94%). The median biologically effective dose (BED) for delivered dose, mean dose, minimum dose and D95% were 51.3 Gy, 54.13 Gy, 29.3 Gy and 51.3 Gy, respectively. The median BED max dose to spinal cord and cauda equine were 59 Gy (15-114) and 68 Gy (7.5-136). There were no grade 3 or higher early or late toxicities reported.

**Conclusion:** In this retrospective analysis of patients who received CsSBRT treatments, we were able to achieve excellent local control safely with minimal toxicity. We recommend at least 2-4 mm space between the target and the spinal cord or cauda equine in order to deliver CsSBRT.

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### Impact of Immunotherapy Timing on Local Control, and Survival in Stereotactic Body Radiation Therapy (SBRT) to the Spine

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**Purpose/Objective(s):** The purpose of the study is to retrospectively investigate if immunotherapy sequencing was associated with local control, fracture risk, or survival in SBRT for spine metastases.

**Materials/Methods:** All patients at our institution who received SBRT and systemic therapy for spine metastases from 2010-2019 were